SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204
Estd:1980

INFORMATION TECHNOLOGY
(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)

III/IV B.TECH

I-SEMESTER

(With effect from **2019-2020**Admitted Batch onwards**)**

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 HS 3101	Managerial Economics and Financial Accountancy	HS	3	3	--	--	25	75	100
B19 IT 3101	Computer Networks	PC	3	3	--	--	25	75	100
B19 IT 3102	Compiler Design	PC	3	3	--	--	25	75	100
B19 IT 3103	Artificial Intelligence	PC	4	3	1	--	25	75	100
B19 IT 3104	Design and Analysis of Algorithms	PC	3	3	--	--	25	75	100
# PE-I	Professional Elective-I	PE	3	3	--	--	25	75	100
B19 IT 3110	Computer Networks & Compiler Design Lab	PC	1.5	--	--	3	20	30	50
B19 IT 3111	AI Tools & Techniques Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3101	Employability Skills – I	MC	--	3	--	--	--	--	--
B19 MC 3103	Advanced Coding	MC	--	--	--	3	--	--	--
		TOTAL	**22**	**21**	**1**	**9**	**190**	**510**	**700**

	Course Code	Course
#PE-I	B19 IT 3105	Fundamentals Of Image Processing
	B19 IT 3106	No SQL Databases
	B19 IT 3107	Scripting Languages
	B19 IT 3108	Computer Graphics
	B19 IT 3109	R-Programming

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B19HS3101	**HS**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY

(Common to CE & IT)

Course Objectives: Students are expected to learn

1.	To Study Managerial Economics and Demand Analysis
2.	To familiarize about the Concepts of Cost and Break-Even Analysis.
3.	To understand the nature of markets and to know the Pricing Policies
4.	To learn about accounting cycle and preparation of Financial Statements.
5.	To know the concept of Capital and sources of raising and Depreciation

Course Outcomes: At the end of the course the student will be able to

S. No	Outcome	KL
1.	Equip oneself with the knowledge of estimating the Demand and demand elasticities for a product.	K2
2.	Have knowledge of Cost and its types and ability to calculate BEP	K3
3.	Understand the nature of different markets	K2
4.	Uunderstand Pricing Practices prevailing in today's business world	K2
5.	Prepare Financial Statements and know how to calculate Profit & Loss for a firm	K3
6.	Know Types of capital and their sources and know how to calculate Depreciation	K2

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Managerial Economics and demand Analysis:** **Managerial Economics:** Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics.**Demand Analysis:** Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of demand forecasting and its Methods.
UNIT-II (10 Hrs)	**Cost Analysis**: Importance of cost analysis, **Types of Cost-** Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; **Elements of costs** – Material, Labour, Expenses; **Methods of costing** - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. **Break-even analysis:** Determination of Breakeven point - Applications, Assumptions and Limitations of Break -even analysis (Theory only).
UNIT-III (10 Hrs)	**Introduction to Markets & Pricing Policies** **Market Structures:** Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. **Pricing:** Importance of pricing and its meaning;**Methods of Pricing**: **Cost Based** -Full cost, Mark-up, Marginal &Break-even; **Demand Based** - Penetrating, Skimming; **Competition Based-** Going rate, Sealed Bid, Discount; **Internet Pricing** - Flat-rate, Usage sensitive.

UNIT-IV (8 Hrs)	**Introduction to Financial Accounting:** Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (outlines only).
UNIT-V (12 Hrs)	**Capital & Depreciation:** Types of Capital - Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, medium term and Long term. **Depreciation -** Meaning, Importance and causes of depreciation; Methods of Depreciation- Straight line and diminishing balancing methods (Theory only)

Text Books:

1.	A R Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N.Appa Rao, Dr.P. Vijayakumar:Managerial Economics and Financial Analysis', Cengage Publications, New Delhi

Reference Books:

1.	Dr.B.Kuberudu& T.V. Ramana : Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K. Gupta & R.K. Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19 IT 3101	PC	3	--	--	3	25	75	3 Hrs.

COMPUTER NETWORKS

(For IT)

Course Objectives: The main objectives of this course are

1.	Study the basic taxonomy and terminology of the computer networking and enumerate the Physical and data link layers of OSI model and TCP/IP model.
2.	Understand Different aspects related to Physical layer such as transmission media, Switching etc
3.	Study data link layer concepts, design issues, protocols and ad-hoc networks.
4.	Understand issues related to error detection , flow control, collision in networks etc.

Course Outcomes: At the end of this course, the student should be able to

S.No	Outcome	KL
1.	Understand components, Data flow in data communication and differentiate layered protocol suits	K2
2.	Differentiate among transmission media, switching networks and their thrust applications	K2
3.	Analyze problems related to error detection, flow control, link control with respect to data link layer	K4
4.	Understand MAC layer protocols and LAN technologies	K2

SYLLABUS

UNIT-I (10 Hrs)	Introduction: Data Communication, components, data representation, dataflow; Networks: network criteria, physical structures, network models, categories of network,inter Connection of networks; The Internet: brief history, internet today, Standard organization, internet standards, Protocol Layering, TCP/IP Protocol Suite, The OSI model.
UNIT-II (10 Hrs)	Physical layer: Data & Signals, Transmission Impairment, Data Rate Limits, Performance, Multiplexing, Spread Spectrum and Transmission Media: Guided Media, Unguided Media, introduction to switching: Circuit Switched Networks, Packet Switching.
UNIT-III (10 Hrs)	Data Link Layer: Introduction, Link layer Addressing, Error Detection andCorrection: Types of Errors, Redundancy, Detection vs Correction, Coding, block coding, cyclic codes: cyclic redundancy check, polynomials, cyclic code analysis, advantages, hard ware implementation, Checksum, Forward Error Correction, DLC Services, Data Link Layer Protocols
UNIT-IV (8 Hrs)	Data Link layer: HDLC: configuration and transfer modes, framing, Point to Point protocol(PPP): services, framing, transition phase, multiplexing Random Access: ALOHA, CSMA, CSMA/CD, CSMA/CA, Controlled Access: Reservation, Polling, Token Passing, Channelization: FDMA, TDMA

UNIT-V (12 Hrs)	Ethernet Protocol, Standard Ethernet, Fast Ethernet, Gigabit Ethernet, 10 Gigabit Ethernet, IEE-802.11: Architecture, MAC sub layer, addressing mechanism, Physical Layer Bluetooth: Architecture, Bluetooth layers.
Text Books:	
1.	Data Communication and Networking , Behrouz A. Forouzan, Mc Graw Hill, 5th Edition, 2012
2.	Computer Networks , Andrew S. Tanenbaum, David J. Wetherall, Pearson Education India; 5 edition, 2013
Reference Books:	
1.	Computer networks, Mayank Dave,CENGAGE.
2.	Computer Networks: A Systems Approach, LL Peterson, BS Davie, Morgan-Kauffman , 5th Edition,2011.
3.	Computer Networking: A Top-Down Approach JF Kurose, KW Ross, Addison-Wesley , 5th Edition,2009
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105183/

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3102	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

COMPILER DESIGN

(For IT)

Course Objectives:

1.	To study the various phases in the design of a compiler
2.	To understand the design of top-down and bottom-up parsers
3.	To understand syntax directed translation schemes
4.	To introduce LEX and YACCtool
5.	To learn to develop algorithms to generate code for a targetmachine

Course Outcomes: At the end of this course, the student should be able to

S.No	Outcome	Knowledge Level
1.	Use LEX and YACC tools for developing a scanner and a parser	K3
2.	Design and implement LL and LR parsers	K3
3.	Design algorithms to perform code optimization in order to improve the performance of a program in terms of space and time complexity	K3
4.	Apply algorithms to generate machine code	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction: Language Processors, the structure of a compiler, the science of building a compiler, programming language basics. Lexical Analysis: The Role of the Lexical Analyzer, Input Buffering, Recognition of Tokens, The Lexical-Analyzer Generator Lex, Finite Automata, From Regular Expressions to Automata, Design of a Lexical-Analyzer Generator, Optimization of DFA-Based Pattern Matchers.
UNIT-II (10 Hrs)	Syntax Analysis: Introduction, Context-Free Grammars, Writing a Grammar, Top-Down Parsing, Recursive and Non recursive top down parsers, Bottom-Up Parsing, Introduction to LR Parsing: Simple LR, More Powerful LR Parsers, Using Ambiguous Grammars, and Parser Generators.
UNIT-III (10 Hrs)	Syntax-Directed Translation: Syntax-Directed Definitions, Evaluation Orders for SDD's, Applications of Syntax-Directed Translation, Syntax-Directed Translation Schemes, and Implementing L-Attributed SDD's. Intermediate-Code Generation: Variants of Syntax Trees, Three-Address Code, Types and Declarations, Type Checking, Control Flow, Back patching, Switch-Statements, Intermediate Code forProcedures.
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UNIT-IV (8 Hrs)	Run-Time Environments: Storage organization, Stack Allocation of Space, Access to Nonlocal Data on the Stack, Heap Management, Introduction to Garbage Collection, Introduction to Trace-Based Collection. Code Generation: Issues in the Design of a Code Generator, The Target Language, Addresses in the Target Code, Basic Blocks and Flow Graphs, Optimization of Basic Blocks, A Simple Code Generator, Peephole Optimization, Register Allocation and Assignment, Dynamic Programming Code-Generation.

UNIT-V (12 Hrs)	Machine-Independent Optimizations: The Principal Sources of Optimization, Introduction to Data-Flow Analysis, Foundations of Data-Flow Analysis, Constant Propagation, Partial-Redundancy Elimination, Loops in Flow Graphs.
Text Books:	
1.	Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman,Pearson.
2.	Compiler Construction-Principles and Practice, Kenneth C Louden, Cengage Learning.
Reference Books:	
1.	Modern compiler implementation in C, Andrew W Appel, Revised edition, Cambridge University Press.
2.	The Theory and Practice of Compiler writing, J. P. Tremblay and P. G. Sorenson,TMH
3.	Writing compilers and interpreters, R. Mak, 3rd edition, Wiley studentedition**.**

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3103	**PC**	**3**	**1**	**--**	**4**	**25**	**75**	**3 Hrs.**

ARTIFICIAL INTELLIGENCE

(For IT)

Course Objectives:

1	To have a basic proficiency in a traditional AI language including an ability to write simple to intermediate programs and an ability to understand code written in that language
2	To understand the basic issues of knowledge representation and blind and heuristic search, as well as an understanding of other topics such as minimax, resolution that play an important role in AI programs
3	To have a basic understanding of some of the more advanced topics of AI

Course Outcomes: At the end of this course, the student should be able to

S.No	Outcome	Knowledge Level
1	Student would able to understand the basic applications of AI and problems that can be solved by AI	K3
2	Student would apply the problem solving strategies to generate best AI solutions using state space search	K3
3	Student would apply AI languages to represent knowledge base	K3
4	Student would apply AI tools to represent knowledge base	K3
5	Student would apply uncertainty techniques to solve AI real time problems	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction, history, intelligent systems, foundations of AI, applications, tic-tac-toe game playing, development of AI languages, current trends.
UNIT-II (10 Hrs)	Problem solving: state-space search and control strategies: Introduction, general problem solving, characteristics of problem, exhaustive searches, heuristic search techniques, iterative deepening A*, constraint satisfaction Problem reduction and game playing: Introduction, problem reduction, game playing, alpha beta pruning.
UNIT-III (10 Hrs)	Logic concepts: Introduction, propositional calculus, proportional logic, natural deduction system, axiomatic system, semantic tableau system in proportional logic, resolution refutation in proportional logic, predicate logic.
UNIT-IV (8 Hrs)	Knowledge representation: Introduction, approaches to knowledge representation, knowledge representation using semantic network, extended semantic networks for KR, knowledge representation using frames Advanced knowledge representation techniques: Introduction, conceptual dependency theory, script structure, CYC theory, case grammars, semantic web.

<table>
<tr><td>UNIT-V
(12 Hrs)</td><td>Expert system and applications: Introduction phases in building expert systems, expert system versus traditional systems
Uncertainty measure: probability theory: Introduction, probability theory, Bayesian belief networks, certainty factor theory, dempster-Shafer theory.
Fuzzy sets and fuzzy logic: Introduction, fuzzy sets, fuzzy set operations, types of membership functions, multi valued logic, fuzzy logic, linguistic variables and hedges, fuzzy propositions, inference rules for fuzzy propositions, fuzzysystems.</td></tr>
<tr><td></td><td></td></tr>
<tr><td colspan="2">Text Books:</td></tr>
<tr><td>1.</td><td>Artificial Intelligence- Saroj Kaushik, CENGAGELearning.</td></tr>
<tr><td>2.</td><td>Artificial intelligence, A modern Approach , 2nded, Stuart Russel, Peter Norvig,PEA.</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td colspan="2">Reference Books:</td></tr>
<tr><td>1.</td><td>Artificial Intelligence- Deepak Khemani, TMH,2013.</td></tr>
<tr><td>2.</td><td>Introduction to Artificial Intelligence, Patterson, PHI.</td></tr>
<tr><td>3.</td><td>Artificial intelligence, structures and Strategies for Complex problem solving, George F Lugar, 5th ed,PEA.</td></tr>
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Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3104	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

DESIGN AND ANALYSIS OF ALGORITHMS

(For IT)

Course Objectives: On completing this course student will be able to

1	Solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, and branch and bound and writing programs for these solutions
2	Analyze the asymptotic performance of algorithms.
3	Demonstrate a familiarity with major algorithms and data structures
4	Synthesize efficient algorithms in common engineering design situations.

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1	Apply the mathematical principle to analyze the efficiency of algorithms by measuring time complexity & space complexity.	K3
2	Apply the Divide-and-Conquer strategy and Greedy Method for solving the complex problems & analyze the performance of solutions.	K3
3	Apply the optimistic strategies Dynamic Programming for computational problems in computer field.	K3
4	Apply the Backtracking and Branch-and-bound strategies for solving complex problems	K3
5	Understand the basic concepts of NP-Hard and NP- Complete and Solve string matching using various algorithms.	K2

SYLLABUS

UNIT-I (10 Hrs)	Introduction: Algorithm Definition, Algorithm Specification, performance Analysis, Randomized Algorithms. Sets & Disjoint set union: introduction, union and find operations. Basic Traversal & Search Techniques: Techniques for Graphs, connected components and Spanning Trees, Bi-connected components and DFS.
UNIT-II (10 Hrs)	Divide and Conquer: General Method, Defective chessboard, Binary Search, finding the maximum and minimum, Merge sort, Quick sort. The Greedy Method: The general Method, container loading, knapsack problem, Job sequencing with deadlines, minimum cost spanning Trees.
UNIT-III (10 Hrs)	Dynamic Programming: The general method, multistage graphs, All pairs-shortest paths, single-source shortest paths: general weights, optimal Binary search trees, 0/1 knapsack, reliability Design, The traveling salesperson problem
UNIT-IV (8 Hrs)	Backtracking: The General Method, The 8-Queens problem, sum of subsets, Graph coloring, Hamiltonian cycles, 0/1 knapsack problem. Branch and Bound: FIFO Branch-and-Bound, LC Branch-and-Bound, 0/1 Knapsack problem, Traveling salesperson problem.

UNIT-V (12 Hrs)	NP-Hard and NP-Complete problems: Basic concepts, Cook's Theorem. String Matching: Introduction, String Matching-Meaning and Application, NaÏve String Matching Algorithm, Rabin-Karp Algorithm, Knuth-Morris-Pratt Automata, Tries, Suffix Tries.

Text Books:	
1.	EllisHorowitz,SartajSahni,SanguthevarRajasekaran,"FundamentalsofComputer Algorithms", 2nd Edition, Universities Press.
2.	Harsh Bhasin, "Algorithms Design & Analysis", Oxford University Press.
Reference Books:	
1.	Horowitz E. Sahani S: "Fundamentals of Computer Algorithms", 2nd Edition, Galgotia Publications,2008.
2.	S. Sridhar, "Design and Analysis of Algorithms", Oxford UniversityPress.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3105	**PE**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

FUNDAMENTALS OF IMAGE PROCESSING

(For IT)

Course Objectives:

1.	Learn Discuss digital image fundamentals digital image fundamentals.
2.	Be exposed to simple image processing techniques.
3.	Be familiar with image compression techniques.
4.	Learn to represent color image in form of features.
5.	Be exposed to segmentation techniques.

Course Outcomes:

S.No	Outcome	Knowledge Level
1.	Discuss digital image fundamentals.	K2
2.	Analyze and apply image enhancement and restoration techniques.	K4
3.	Analyze and apply image compression techniques.	K4
4.	Distinguish between different features of color images.	K3
5.	Apply image segmentation techniques.	K3

SYLLABUS

UNIT-I (8 Hrs)	DIGITAL IMAGE FUNDAMENTALS Introduction – Origin – Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels.
UNIT-II (10 Hrs)	IMAGE ENHACEMENT Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening of Spatial Filtering – Frequency Domain: Introduction to Fourier Transform – Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters.
UNIT-III (9 Hrs)	IMAGE RESTORATION Degradation model, properties, Noise models – Mean Filters – Order Statistics – Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters, Weiner Filtering .
UNIT-IV (9 Hrs)	IMAGE COMPRESSION Need for Data Compression, Compression: Fundamentals – Huffman , Run length encoding, Arithmetic coding, Compression Standards - JPEG
UNIT-V (10 Hrs)	COLOR IMAGE PROCESSING AND SEGMENTATION Color fundamentals, color models, the RGB, the CMY and CMYK, the HSI color models, color transformations, color slicing, tone and color corrections, histogram processing, segmentation fundamentals, and thresholding.

Text Books:

1.	Rafael C. Gonzales, Richard E. Woods, “Digital Image Processing”, Third Edition, Pearson Education, 2010

2.	Anil Jain K. “Fundamentals of Digital Image Processing”, PHI Learning Pvt. Ltd., 2011.
Reference Books:	
1.	Kenneth R. Castleman, Digital Image Processing, Pearson, 2006.
2.	Malay K. Pakhira, “Digital Image Processing and Pattern Recognition”, First Edition, PHI Learning Pvt. Ltd., 2011.
3.	Willliam K Pratt, “Digital Image Processing”, John Willey, 2002.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3106	PE	3	--	--	3	25	75	3 Hrs.

NoSQL DATABASES

(For IT)

Course Objectives: The objective of the course is to:

1.	Define, compare, and use the four types of NoSQL Databases (Document-oriented, Key Value Pairs, Column oriented and Graph)
2.	Demonstrate an understanding of the detailed architecture, define objects, load data, query data and performance tune Column-oriented NoSQL databases
3.	Explain the detailed architecture, define objects, load data, query data and performance tune Document oriented NoSQL databases
4.	Ability to design entity relationship model and convert entity relationship diagrams into RDBMS and formulate SQL queries on the data

Course Outcomes: After the completion of the course, student will be able to do

S.No	Outcome	Knowledge Level
1.	Identify what type of NoSQL database to implement based on business requirements (key-value, document, full text, graph,etc.)	K3
2.	Apply NoSQL data modeling from application specific queries	K4
3.	Use Atomic Aggregates and denormalization as data modelling techniques to optimize query processing	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction to NoSQL: Definition And Introduction, Sorted Ordered Column-Oriented Stores, Key/Value Stores, Document Databases, Graph Databases, Examining Two Simple Examples, Location Preferences Store, Car Make And Model Database, Working With Language Bindings.
UNIT-II (10 Hrs)	Interacting with NoSQL: If NoSql Then What, Language Bindings For NoSQL Data Stores, Performing Crud Operations, Creating Records, Accessing Data, Updating And Deleting Data
UNIT-III (10 Hrs)	NoSQL Storage Architecture: Working With Column-Oriented Databases, Hbase Distributed Storage Architecture, Document Store Internals, Understanding Key/Value Stores In Memcached And Redis, Eventually Consistent Non-Relational Databases.
UNIT-IV (8 Hrs)	NoSQL Stores: Similarities Between Sql And Mongodb Query Features, Accessing Data From Column- Oriented Databases Like Hbase, Querying Redis Data Stores, Changing Document Databases, Schema Evolution In Column-Oriented Databases, Hbase Data Import And Export, Data Evolution In Key/ValueStores.

UNIT-V (12 Hrs)	Indexing and Ordering Data Sets : Essential Concepts Behind A Database Index, Indexing And Ordering In Mongodb, Creating and Using Indexes In Mongodb, Indexing And Ordering In Couchdb, Indexing In Apache Cassandra.
Text Books:	
1.	Pramod Sadalage and Martin Fowler, NoSQL Distilled, Addison-Wesley Professional,2012.
2.	Dan McCreary and Ann Kelly, Making Sense of NoSQL, Manning Publications,2013.
Reference Books:	
1.	Shashank Tiwari, Professional NoSQL, Wrox Press, Wiley, 2011, ISBN:978-0-470-94224-6
2.	Gaurav Vaish, Getting Started with NoSQL, Packt Publishing,2013.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3107	**PE**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

SCRIPTING LANGUAGES

(For IT)

Course Objectives: From the course the student will

1.	Understand the concepts of scripting languages for developing web based projects
2.	Illustrates object-oriented concepts like PHP, PYTHON,PERL
3.	Create database connections using PHP and build the website for the world
4.	Demonstrate IP address for connecting the webservers
5.	Analyze the internet ware application, security issues and frame works for application

Course Outcomes: After the completion of the course, student will be able to do the following

S.No	Outcome	Knowledge Level
1.	Ability to understand the PERL scripting languages	K2
2.	Understand the fundamentals of PHP to develop secured web application.	K3
3.	Explain syntax and variables in TCL	K2
4.	Master an understanding of python especially the object-oriented concepts	K2

SYLLABUS

UNIT-I (10 Hrs)	Introduction to PERL and Scripting: Scripts and Programs, Origin of Scripting , Scripting Today, Characteristics of Scripting Languages, Uses for Scripting Languages, Web Scripting, and the universe of Scripting Languages. PERL- Names and Values, Variables, Scalar Expressions, Control Structures, arrays, list, hashes, strings, pattern and regular expressions, subroutines.
UNIT-II (10 Hrs)	Advanced PERL: Finer points of looping, pack and unpack, file system, eval, data structures, packages, modules, objects, interfacing to the operating system, Creating Internet ware applications, Dirty Hands Internet Programming, securityIssues. PHP Basics: PHP Basics- Features, Embedding PHP Code in your Web pages, Outputting the data to the browser, Data types, Variables, Constants, expressions, string interpolation, control structures, Function, Creating a Function, Function Libraries, Arrays, strings and Regular Expressions.
UNIT-III (10 Hrs)	Advanced PHP Programming: PHP and Web Forms, Files, PHP Authentication and Methodologies- Hard Coded, File Based, Database Based, IP Based, Login Administration, Uploading Files with PHP, Sending Email using PHP, PHP Encryption Functions, the Mcrypt package, Building Web sites for the World.
UNIT-IV (8 Hrs)	TCL: TCL Structure, syntax, Variables and Data in TCL, Control Flow, Data Structures, input/output, procedures , strings , patterns, files, Advance TCL- eval, source, exec and up level commands, Name spaces, trapping errors, event driven programs, making applications internet aware, Nuts and Bolts Internet Programming, Security Issues, C Interface. Tk-Visual Tool Kits, Fundamental Concepts of Tk, Tk by example, Events and Binding ,Perl-Tk.

UNIT-V (12 Hrs)	Python: Introduction to Python language, python-syntax, statements, functions, Built-in-functions and Methods, Modules in python, Exception Handling. Integrated Web Applications in Python – Building Small, Efficient Python Web Systems, Web Application Framework.

Text Books:	
1.	The World of Scripting Languages, David Barron, Wiley Publications.
2.	Python Web Programming, Steve Holden and David Beazley, New Riders Publications.
3.	Beginning PHP and MySQL, 3rd Edition, Jason Gilmore, Apress Publications (Dreamtech).

Reference Books:	
1.	Open Source Web Development with LAMP using Linux, Apache, MySQL, Perl and PHP, J.Lee and B.Ware (Addison Wesley) Pearson Education. ProgrammingPython,M.Lutz,SPD.
2.	PHP 6 Fast and Easy Web Development, Julie Meloni and Matt Telles, Cengage Learning Publications.
3.	Tcl and the Tk Tool kit, Ousterhout, PearsonEducation.
4.	PHP and MySQL by Example, E.Quigley, Prentice Hall(Pearson).
5.	Perl Power, J.P.Flynt, CengageLearning.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3108	PE	3	--	--	3	25	75	3 Hrs.

COMPUTER GRAPHICS

(For IT)

Course Objectives: From the course the student will be able

1.	To develop, design and implement two- and three-dimensional graphical structures.
2.	To enable students to acquire knowledge Multimedia compression and animations.
3.	To learn Creation, basic primitives using OPENGL and learn methods for rendering and retracing.

Course Outcomes: After learning the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	Illustrate and apply the basics of computer graphics, different graphics systems and applications of computer graphics with various algorithms for line, circle and ellipse drawing objects.	K3
2.	Apply and analyze 2D transformations and perform clipping.	K4
3.	Apply and analyze projections and perform 3D transformations	K4
4.	Illustrate different graphic color models and Basic programming in OPENGL	K3
5.	Illustrate different shading models, rendering objects and understand basics of ray tracing.	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction to Graphics: Application areas of Computer Graphics, overview of graphics systems, video- display devices, graphics monitors and work stations and input devices **2D Primitives:** Output primitives – Line, Circle and Ellipse drawing algorithms, Attributes of output primitives,
UNIT-II (10 Hrs)	**2D Transformations** :Two dimensional Geometric transformations, Two dimensional viewing – Line, Polygon, Curve and Text clipping algorithms
UNIT-III (10 Hrs)	**3D Concepts:** Parallel and Perspective projections, Three dimensional object representation– Polygons, Curved lines, Splines, Quadric Surfaces, Visualization of data sets, 3D transformations, Viewing.
UNIT-IV (8 Hrs)	**Graphics Programming**: Color Models – RGB, YIQ, CMY, HSV, Animations – General Computer Animation, Raster, Keyframe. Graphics programming using OPENGL – Basic graphics primitives, Drawing three dimensional objects, Drawing three dimensional scenes
UNIT-V (12 Hrs)	**Rendering:** Introduction to shading models, Flat and Smooth shading, Adding texture to faces, Adding shadows of objects, Building a camera in a program, Creating shaded objects. Basics of Ray Tracing.

Text Books:

1.	Donald Hearn, Pauline Baker, Computer Graphics – C Version, second edition, Pearson Education,2004.

2.	Schaum's Outline of Computer Graphics Second Edition, Zhigang Xiang, Roy A. Plastock.
Reference Books:	
1.	James D. Foley, Andries Van Dam, Steven K. Feiner, John F. Hughes, Computer Graphics- Principles and practice, Second Edition in C, Pearson Education,2007.
2.	F.S. Hill, Computer Graphics using OPENGL, Second edition, Pearson Education,2003

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3109	PE	3	--	--	3	25	75	3 Hrs.

R-PROGRAMMING

(For IT)

Course Objectives: From the course the student will be able

1.	To be familiar with basic techniques of R.
2.	Solve problems using statistical methods present in R.
3.	Use different types of functions available in R.
4.	Create different types of graphs and plots.
5.	To practice different Regression techniques.

Course Outcomes: After learning the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	Identify the basic data types and advanced data structures in R	K3
2.	Develop user defined functions and implement control statements	K3
3.	Analyze probability, linear algebra operations ,statistical distributions and graphs	K4
4.	Determine distribution techniques and linear models using regression	K4

SYLLABUS

UNIT-I (10 Hrs)	Introduction, How to run R, R Sessions and Functions, Basic Math, Variables, Data Types, Vectors, Conclusion, Advanced Data Structures, Data Frames, Lists, Matrices, Arrays, Classes.
UNIT-II (10 Hrs)	R Programming Structures, Control Statements, Loops, - Looping Over Non vector Sets,- If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion, A Quick sort Implementation-Extended Extended Example: A Binary Search Tree.
UNIT-III (10 Hrs)	Doing Math and Simulation in R, Math Function, Extended Example Calculating Probability- Cumulative Sums and Products-Minima and Maxima- Calculus, Sorting, Linear Algebra Operation on Vectors and Matrices, Extended Example: Vector cross Product- Extended Example: Finding Stationary Distribution of Markov Chains, Set Operation, Input /out put, Accessing the Keyboard and Monitor, Reading and writer Files.
UNIT-IV (8 Hrs)	Graphics, Creating Graphs, The Workhorse of R Base Graphics- the plot() , points() , legend() , text(), locator(), barplot(), boxplot(), Histogram - hist(),PIE-CHART: pie(),line(), abline() functions, Customizing Graphs, Saving Graphs to Files.
UNIT-V (12 Hrs)	Probability Distributions, Normal Distribution- Binomial Distribution- Poisson Distributions Other Distribution, Basic Statistics, Correlation and Covariance, T-Tests,- ANOVA. Linear Models, Simple Linear Regression, -Multiple Regression Generalized Linear Models, Logistic Regression, - Poisson Regression- other Generalized Linear Models-Survival Analysis, Nonlinear Models, Splines- Decision- Random Forests,

Text Books:	
1.	R for Everyone, Lander, Pearson
2.	The Art of R Programming, Norman Matloff, Cengage Learning
Reference Books:	
1.	R Cookbook, PaulTeetor, Oreilly.
2.	R in Action,RobKabacoff, Manning
e-Resources	
1.	https://www.tutorialspoint.com/r/index.htm

Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3110	**PC**	**--**	**--**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

COMPUTER NETWORKS & COMPILER DESIGN LAB

Course Objectives:

1	To learn and use various network commands
2	To learn various error detection and correction methods
3	To implement and analyze various data link layer protocols
4	To implement various parsers

Course Outcomes

S.No	Outcome	Knowledge Level
1	Able to use various protocols commands for network establishment	K2
2	Able to Implement error correction codes for correct data transmission in data link layer	K3
3	Able to Implement scanners ,parsers, for lexical & syntax analysis	K3
4	Able to implement code optimizers & code generators in compilers	K3

LIST OF EXPERIMENTS

1) Connect the computers in Local Area Network
2) Learn to use commands like tcpdump, netstat, ifconfig, nslookup and trace route.
3) Implement Data Link Framing method - Character Count.
4) Implement Data link framing method - Bit stuffing and Destuffing.
5) Implement Error detection method - even and odd parity.
6) Implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
7) Implement Data Link protocols - Unrestricted simplex protocol
8) Implement data link protocols - Stop and Wait protocol
9) Simulate error correction code (like CRC).
10) Write a C program to recognize strings under _a‘, _a*b+‘, _abb‘.
11) Write a C program to test whether a given identifier is valid or not.
12) Write a C program to simulate lexical analyser for validating operators
13) Write a C program for constructing recursive descent parsing.
14) Write a C program to implement LALR parsing.
15) Write a C program to implement operator precedence parsing.

Reference Books:

1.	Data Communication and Networking , Behrouz A. Forouzan, Mc Graw Hill, 5th Edition, 2012
2	Computer Networks , Andrew S. Tanenbaum, David J. Wetherall, Pearson Education India; 5 edition, 2013
3	Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman, Pearson.
4	Compiler Construction-Principles and Practice, Kenneth C Louden, Cengage Learning.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3111	**PC**	**--**	**--**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

AI TOOLS & TECHNIQUES LAB

(For IT)

Course Objectives:

1	Study the concepts of Artificial Intelligence
2	Learn the methods of solving problems using Artificial Intelligence
3	Introduce the concepts of machine learning

Course Outcomes: At the end of the course, the students will be able to:

S.No	Outcome	Knowledge Level
1	Student would analyze problems that are amenable to solution by AI method	K4
2	Student would Identify appropriate AI methods to solve a given problem	K4
3	Student would Use language/framework of different AI methods for solving problems	K4
4	Student would Implement basic AI algorithms Design and carry out an empirical evaluation of different algorithms on problem formalization, and state the conclusions that the evaluation supports	K4

LIST OF EXPERIMENTS

1. Study of Prolog.
2. Write simple fact for the statements using PROLOG.
3. Write predicates One converts centigrade temperatures to Fahrenheit, the other checks if a temperature is below freezing
4. Write a program to solve the Monkey Banana problem.
5. Write a program in turbo prolog for medical diagnosis and show the advantage and disadvantage of green and red cuts
6. Write a program to implement factorial, Fibonacci of a given number
7. Implementation of Towers of Hanoi Problem using PROLOG
8. Write a program to solve 4-Queen and 8-puzzleproblem.
9. Implementation of DFS and BFS for water jug problem using PROLOG
10. Write a program to solve traveling salesman problem.
11. Write a program to solve water jug problem using PROLOG
12. Implementation of A* Algorithm using PROLOG
13. Implementation of Hill Climbing Algorithm using PROLOG

Reference books:

1	Artificial Intelligence- Saroj Kaushik, CENGAGELearning.
2	Artificial intelligence, A modern Approach , 2nded, Stuart Russel, Peter Norvig,PEA.
3	www.tutorialpoint.prolog

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3101	MC	3	--	--	--	--	--	--

EMPLOYABILITY SKILLS - I

Common to All Branches

Part-A: Verbal and Soft Skills-I

Course Objectives:

1.	To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2.	To familiarize the learner with high frequency words as they would be used in their professional career.
3.	To inculcate logical thinking in order to frame and use data as per the requirement
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5.	To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes: The students will be able to

S.No	Outcome	Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.	K3
2	Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.	K3
3.	Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.	K3
4.	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.	K3
5.	Apply soft skills in the work place and build better personal and professional relationships making informed decisions.	K3

SYLLABUS

UNIT-I	**Grammar: (VA)** Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns- number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.
UNIT-II	**Vocabulary: (VA)** Synonyms and synonym variants (with emphasis on high frequency words), antonyms and antonym variants (with emphasis on high frequency words), contextual meanings with regard

	to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.
UNIT-III	**Reasoning: (VA)** Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms.
UNIT-IV	**Usage: (VA)** Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.
UNIT-V	**Soft Skills:** Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.
Text Books:	
1.	Oxford Learners„s Grammar – Finder by John Eastwood, Oxford Publication.
2.	R S Agarwal„s books on objective English and verbal reasoning
3.	English Vocabulary in Use- Advanced , Cambridge University Press
4.	Collocations In Use, Cambridge University Press
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi
Reference Books:	
1.	English Grammar in Use by Raymond Murphy, CUP
2.	Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from ‗IMS, Career Launcher and Time„ institutes for competitive exams
4.	The Art of Public Speaking by Dale Carnegie
5.	The Leader in You by Dale Carnegie
6.	Emotional Intelligence by Daniel Golman
7.	Stay Hungry Stay Foolish by Rashmi Bansal
8.	I have a Dream by Rashmi Bansal.

Part-B: Quantitative Aptitude-I		
Course Objectives:		
1.	To familiarize students with basic problems on numbers and ratios problems.	
2.	To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.	
3.	To enable the students to work efficiently on percentage values related to shares, profit and loss problems.	
4.	To inculcate logical thinking by exposing the students to reasoning related questions.	
5.	To inculcate logical thinking by exposing the students to reasoning related questions.	
Course Outcomes:		
S. No.	**Course Outcome**	**Knowledge Level**
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	Accurate solving problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies	K3
SYLLABUS		
UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion	
UNIT-II	Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.	
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.	

UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal
3.	Puzzles to puzzle you by shakunataladevi.
References:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for cat by arunsharma.
5.	Elementary and Higher algebra by HS Hall and SR knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3103	MC	--	--	3	0	--	--	--

ADVANCED CODING
(Common to CSE & IT)

Course Objectives:

1.	To understand the basics of Java Packages
2	To learn about Collection Framework
3	To investigate different methods of GUI Programming.
4	To learn about Java collections and Libraries

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1.	Able to solve problems using java collection framework and I/o classes.	K3
2.	Able to develop multithreaded applications with synchronization.	K3
3.	Able to develop applets for web applications	K3
4.	Able to design GUI based applications	K3

SYLLABUS

UNIT-I (10 Hrs)	Standard Library templates and Java collections Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.
UNIT-II (10 Hrs)	The Collections Framework (java.util)- Collections overview, Collection Interfaces, The Collection classes- Array List, Linked List, Hash Set, Tree Set, Priority Queue, Array Deque. Accessing a Collection via an Iterator, Using an Iterator, The For-Each alternative, Map Interfaces and Classes, Comparators, Collection algorithms, Arrays, The Legacy Classes and Interfaces- Dictionary, Hashtable ,Properties, Stack, Vector.
UNIT-III (10 Hrs)	More Utility classes, String Tokenizer, Bit Set, Date, Calendar, Random, Formatter, Scanner.
UNIT-IV (8 Hrs)	GUI Programming with Swing – Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout. Event Handling- The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events.
UNIT-V (12 Hrs)	Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application, Applets – Applets and HTML, Security Issues, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList, JCombo Box, Swing Menus, Dialogs.

References:	
1.	Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt. Ltd.
2.	Understanding Object-Oriented Programming with Java, updated edition, T. Budd, Pearson Education
3	An Introduction to programming and OO design using Java, J. Nino and F.A. Hosch, John Wiley & sons
4	Introduction to Java programming, Y. Daniel Liang, Pearson Education.
5	Object Oriented Programming through Java, P. Radha Krishna, University Press.
6	Programming in Java S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press.
7	Java Programming and Object-oriented Application Development, R. A. Johnson, Cengage Learning.
8	https://www.geeksforgeeks.org/
9	https://www.tutorialspoint.com/

INFORMATION TECHNOLOGY

(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

II-SEMESTER

(With effect from **2019-2020**Admitted Batch onwards**)**

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 IT 3201	Data Warehousing and Data Mining	PC	3	3	--	--	25	75	100
B19 IT 3202	Web Technologies	PC	3	3	--	--	25	75	100
B19 IT 3203	Advanced Computer Networks	PC	3	3	--	--	25	75	100
B19 IT 3204	MOOCs-I	PE	3	3	--	--	25	75	100
# OE-I	Open Elective-I	OE	3	3	--	--	25	75	100
B19 IT 3205	Web Technologies Lab	PC	1.5	--	--	3	20	30	50
B19 IT 3206	Data Mining Lab	PC	1.5	--	--	3	20	30	50
B19 IT 3207	Industrial Training / Skill Development Programs / Research Project in higher learning institutes	PR	1	--	--	--	--	50	50
B19 MC 3201	Employability Skills - II	MC	--	3	--	--	--	--	--
B19 MC 3204	Competitive Coding	MC	--	--	--	3	--	--	--
		TOTAL	**19**	**18**	**--**	**9**	**165**	**485**	**650**

#OE-I	Student has to study one Open Elective offered by CE or ECE or EEE or ME or EM &H from the list enclosed.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19 IT 3201	PC	3	--	--	3	25	75	3 Hrs.

DATA WAREHOUSING AND DATA MINING

(For IT)

Course Objectives:

1.	To understand data warehouse concepts, architecture, business analysis and tools
2.	To understand data pre-processing and data visualization techniques
3.	To study algorithms for finding hidden and interesting patterns in data
4.	To understand and apply various classification and clustering techniques using tools

Course Outcomes: At the end of the course, the students will be able to:

S No	Outcome	Knowledge Level
1.	Design a Data warehouse system and perform business analysis with OLAP tools	K3
2.	Apply suitable pre-processing and visualization techniques for data analysis	K4
3.	Apply frequent pattern and association rule mining techniques for data analysis	K4
4.	Apply appropriate classification techniques for data analysis	K4
5.	Apply appropriate clustering techniques for data analysis	K4

SYLLABUS

UNIT-I (10 Hrs)	Data Warehousing, Business Analysis and On-Line Analytical Processing (OLAP): Basic Concepts, Data Warehousing Components, Building a Data Warehouse, Database Architectures for Parallel Processing, Parallel DBMS Vendors, Multidimensional Data Model, Data Warehouse Schemas for Decision Support, Concept Hierarchies, Characteristics of OLAP Systems, Typical OLAP Operations, OLAP andOLTP.
UNIT-II (10 Hrs)	**Introduction to Data Mining Systems**: Knowledge Discovery Process, Data Mining Techniques, Issues, Applications, Data Objects and Attribute types, Statistical Description of data, Data Visualization, Data similarity and dissimilarity measures, Data Preprocessing – Cleaning, Integration, Reduction, Transformation and discretization.
UNIT-III (10 Hrs)	**Mining Frequent Patterns, Associations and Correlations**: Basic Concepts, Frequent Item set Mining Methods, Pattern Evaluation Methods; Pattern Mining in Multilevel, Multi-Dimensional Space – Constraint Based Frequent Pattern Mining.
UNIT-IV (8 Hrs)	**Classification**: Basic Concepts, Decision Tree Induction, Bayesian Classification, Rule Based Classification, Classification by Back Propagation, Support Vector Machines, Classification using Frequent Patterns, Lazy Learners, Model Evaluation and Selection, Techniques to improve Classification Accuracy.
UNIT-V (12 Hrs)	**Clustering**: Cluster analysis, Partitioning Methods, Hierarchical methods, Density Based Methods, Grid Based Methods, Evaluation of Clustering, Clustering High Dimensional Data, Clustering with Constraints, Outlier Analysis and Outlier Detection Methods.

Text Books:	
1.	Jiawei Han and Micheline Kamber, “Data Mining Concepts and Techniques”, Third Edition, Elsevier,2012.
2.	Pang-Ning Tan, Michael Steinbach and Vipin Kumar, Introduction to Data Mining, Pearson,2016.
Reference Books:	
1.	Alex Berson and Stephen J.Smith, ―Data Warehousing, Data Mining & OLAP‖, Tata McGraw – Hill Edition, 35th Reprint 2016.
2.	K.P. Soman, ShyamDiwakar and V. Ajay, ―Insight into Data Mining Theory and Practice‖, Eastern Economy Edition, Prentice Hall of India,2006.
3.	Ian H.Witten and Eibe Frank, ―Data Mining: Practical Machine Learning Tools and Techniques‖, Elsevier, SecondEdition.
e-Resources	
1.	https://www.saedsayad.com/data_mining_map.htm
2.	https://nptel.ac.in/courses/106/105/106105174/
3.	(NPTEL course by Prof.Pabitra Mitra)http://onlinecourses.nptel.ac.in/noc17_mg24/preview
4	(NPTEL course by Dr. Nandan Sudarshanam & Dr.BalaramanRavindran) http://www.saedsayad.com/data_mining_map.htm

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3202	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

WEB TECHNOLOGIES

(For IT)

Course Objectives: From the course the student will learn

1	Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client
2	Write backend code in PHP language and writing optimized front end code HTML and JavaScript
3	Understand, create, and debug database related queries and Create test code to validate the applications against client requirement
4	Monitor the performance of web applications & infrastructure and Troubleshooting web application with a fast and accurate a resolution

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	**Outcome**	KL
1	Illustrate the basic concepts of HTML and CSS & apply those concepts to design static web page	K1
2	Identify and understand various concepts related to dynamic web pages and validate them using JavaScript	K2
3	Outline the concepts of Extensible markup language &AJAX	K2
4	Create web Applications using Scripting Languages &Frameworks	K3
5	Create and deploy secure, usable database driven web applications using PHP and RUBY	K3

SYLLABUS

UNIT-I (10 Hrs)	HTML: Basic Syntax, Standard HTML Document Structure, Basic Text Markup, Html styles, Elements, Attributes, Heading, Layouts, Html media, Iframes Images, Hypertext Links, Lists, Tables, Forms, GET and POST method, HTML 5, Dynamic HTML. CSS: Cascading style sheets, Levels of Style Sheets, Style Specification Formats, Selector Forms, The Box Model, Conflict Resolution, CSS3.
UNIT-II (10 Hrs)	JavaScript - Introduction to JavaScript, Objects, Primitives Operations and Expressions, Control Statements, Arrays, Functions, Constructors, Pattern Matching using Regular Expressions, Fundamentals of Angular JS and NODE JS Angular Java Script-Introduction to Angular JS Expressions: ARRAY, Objects, Strings, Angular JS Form Validation &Form Submission. Node.js- Introduction, Advantages, Node.js Process Model, Node JS Modules, Node JS File system, Node JS URL module, Node JS Events.
UNIT-III (10 Hrs)	Working with XML: Document type Definition (DTD), XML schemas, XSLT, Document object model, Parsers - DOM and SAX. AJAX A New Approach: Introduction to AJAX, Basics of AJAX, XML Http Request Object, AJAX UI tags, Integrating PHP and AJAX.

UNIT-IV (8 Hrs)	PHP Programming: Introduction to PHP, Creating PHP script, Running PHP script. Working with variables and constants: Using variables, Using constants, Data types, Operators. Controlling program flow: Conditional statements, Control statements, Arrays, functions. Working with forms and Databases such as MySQL, MongoDB
UNIT-V (12 Hrs)	Web Servers- IIS (XAMPP, LAMP) and Tomcat Servers Web development frameworks – Introduction to Ruby, Ruby Scripting, Ruby on rails, Design, Implementation and Maintenance aspects.

Text Books:	
1.	Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson,2013.
2.	Web Technologies, 1st Edition 7th impression, Uttam K Roy, Oxford,2012.
3	Pro Mean Stack Development, 1st Edition, ELadElrom, Apress O'Reilly,2016
4	Java Script & jQuery the missing manual, 2nd Edition, David sawyer mcfarland, O'Reilly,2011.
5	Web Hosting for Dummies, 1st Edition, Peter Pollock, John Wiley & Sons,2013
6	RESTful web services, 1st Edition, Leonard Richardson, Ruby, O'Reilly,2007
Reference Books:	
1.	Ruby on Rails Up and Running, Lightning fast Web development, 1st Edition, Bruce Tate, Curt Hibbs, Oreilly,2006.
2.	Programming Perl, 4th Edition, Tom Christiansen, Jonathan Orwant, O'Reilly,2012.
3.	Web Technologies, HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black book, 1st Edition, Dream Tech,2009.
4.	An Introduction to Web Design, Programming,1st Edition, Paul S Wang, Sanda S Katila, Cengage Learning,2003.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3203	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

ADVANCED COMPUTER NETWORKS

(For IT)

Course Objectives: This course is aimed at enabling the students to

1.	Gain core knowledge of Network layer routing protocols and IP addressing.
2.	Study Transport layer services, and protocols.
3.	Acquire knowledge of Application layer and protocols.
4.	To understand routing issues and advance network layer protocols
5.	To develop some familiarity with current research problems and research methods in advance computer networks.

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1.	Illustrate reference models with layers, protocols, and interfaces	**K2**
2.	Analyze and apply the routing algorithms, Sub netting and Addressing of IP V4andIPV6	**K4**
3.	Describe and Analysis of basic protocols of computer networks, and how they can be used to assist in network design and implementation	**K4**
4.	Illustrate the concepts of WWW and different application layer protocols	**K2**

SYLLABUS

UNIT-I (10 Hrs)	Network layer: Network Layer Services, Packet Switching, Performance, implementation connectionless services, implementation connection-oriented services, comparison of virtual –circuit and datagram subnets. IPV4 Address, Forwarding of IP Packets, Internet Protocol, ICMP v4.
UNIT-II (10 Hrs)	Routing Algorithms–Distance Vector routing, Link State Routing, Path Vector Routing, Unicast Routing Protocol- Internet Structure, Routing Information Protocol, Open-Source Path First, Border Gateway Protocol V4, Broadcast routing, Multicasting routing, Multicasting Basics.
UNIT-III (10 Hrs)	IPv6 Addressing, IPv6 Protocol, Transition from IPv4 to IPv6. Transport Layer Services, connectionless versus connection-oriented protocols. Transport Layer Protocols: UDP: User datagram, Services, Applications. TCP: TCP services, TCP features, segment, A TCP connection, Flow control, error control, congestion control.
UNIT-IV (8 Hrs)	SCTP: SCTP services SCTP features, packet format, An SCTP association, flow control, error control. QUALITY OF SERVICE: flow characteristics, flow control to improve QOS: scheduling, traffic shaping, resource reservation, admission control.
UNIT-V (8 Hrs)	WWW and HTTP, SMTP, MIME, TFTP, FTP, Telnet, Domain name system.

Text Books:	
1.	Data Communication and Networking , Behrouz A. Forouzan, McGraw Hill, 5th Edition,2012
2.	Computer Networks , Andrew S. Tanenbaum, David J. Wetherall, Pearson Education India; 5 edition,2013.
Reference Books:	
1.	Computer networks, Mayank Dave,CENGAGE.
2.	Computer Networks: A Systems Approach , LL Peterson, BS Davie, Morgan-Kauffman , 5th Edition,2011.
3.	Computer Networking: A Top-Down Approach JF Kurose, KW Ross, Addison-Wesley , 5th Edition,2009.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105183/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19 IT 3204	**PE**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

MOOCs-I

Professional Electives-II

(For IT)

MOOCs-I course under Professional Elective –II should belong to the B.Tech. Programme concerned and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration.

The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.

Code	Category	L	T	P	C	I.M	E.M	Exam
#OE-I	**OE**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

OPEN ELECTIVE-I

(For IT)

Student has to study one Open Elective offered by CE or ECE or EEE or ME or EM&H from the list enclosed.

Offered by	Course Code	Course Name	Offered to
CIVIL ENGINEERING	B19CEOE01	Disaster Management	**CSE,ECE,EEE, IT &ME**
	B19CEOE02	Remote Sensing &GIS	
ELECTRONICS & COMMUNICATION ENGINEERING	B19ECOE01	Basic Electronics	**CE,CSE,EEE, IT &ME**
	B19ECOE02	Signals & Systems	
ELECTRICAL & ELECTRONICS ENGINEERING	B19EEOE01	Introduction To Electrical Systems	**CE,CSE,ECE, IT &ME**
	B19EEOE02	Electrical Estimation and Costing	
MECHNAICAL ENGINEERING	B19MEOE01	Operations Research	**CE,CSE,ECE, EEE & IT**
	B19MEOE02	Operations Management	
	B19MEOE03	Total Quality Management	
MATHEMATICS & HUMANITIES	B19BSOE01	Computational statistics with R	**ALL BRANCHES**
	B19BSOE02	Fuzzy Sets and Fuzzy Logic	

Sub Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3205	**PC**	**-**	**--**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

WEB TECHNOLOGIES LAB

(For IT)

Course Objectives: On completing this course student will be able to

1	To acquire knowledge to develop web applications using Java Script ,CSS and XML
2	Ability to develop dynamic web content using PHP
3	To understand Data base connections with PHP
4	To understand the design and development process of a complete web application

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1	Students will be able to develop static web sites using CSS and Java Scripts	K4
2	To implement XML and XSLT for web applications	K3
3	Develop Dynamic web content using PHP	K3
4	To Implement database connections with Mysql and PHP to develop dynamic WebPages	K4

SYLLABUS

1

Design the following static web pages required for an online book store web site.

HOME PAGE:

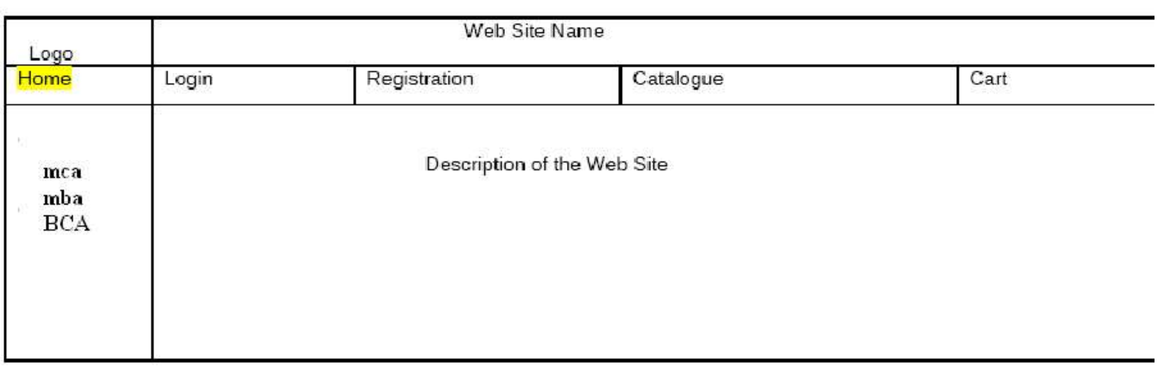
The static home page must contain three **frames**.

Top frame: Logo and the college name and links to Home page, Login page, Registration page, Catalogue page and Cart page (the description of these pages will be given below).

Left frame: At least four links for navigation, which will display the catalogue of respective links.

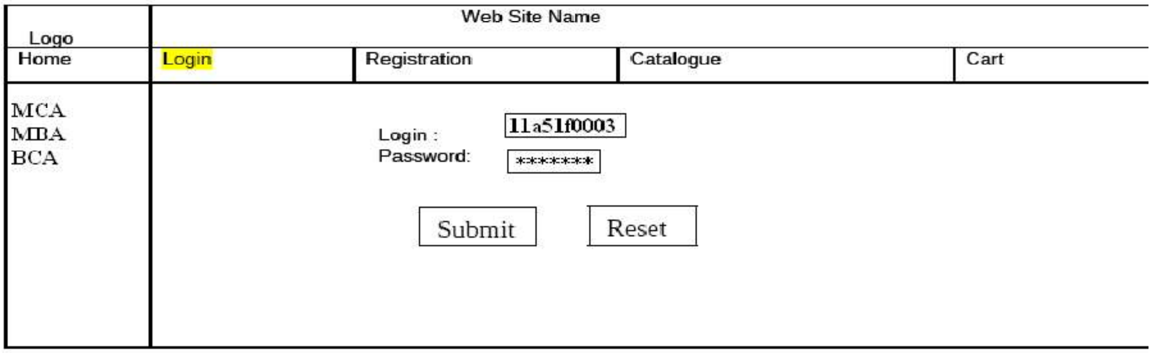
For e.g.: When you click the link **"MCA"** the catalogue for MCA Books should be displayed in the Right frame.

Right frame: The *pages to the links in the left frame must be loaded here*. Initially this page contains description of the web site.



2

LOGIN PAGE:

	
3	**CATOLOGUE PAGE:** The catalogue page should contain the details of all the books available in the web site in a table. The details should contain the following: 1. Snap shot of Cover Page. 2. Author Name. 3. Publisher. 4. Price. 5. Add to cart button.
4	**REGISTRATION PAGE:** Create a "*registration form* "with the following fields 1) Name (Text field) 2) Password (password field) 3) E-mail id (text field) 4) Phone number (text field) 5) Sex (radio button) 6) Date of birth (3 select boxes) 7) Languages known (check boxes – English, Telugu, Hindi, Tamil) 8) Address (text area)
5	**DESIGN A WEB PAGE USING CSS (C**ascading **S**tyle **S**heets**)** which includes the following: 1) Use different font, styles: In the style definition you define how each selector should work (font, color etc.). Then, in the body of your pages, you refer to these selectors to activate the styles
6	**WRITE AN XML** file which will display the Book information which includes the following: 1) Title of the book 2) Author Name 3) ISBN number 4) Publisher name 5) Edition 6) Price Write a Document Type Definition (DTD) to validate the above XML file.
7	Write Ruby program reads a number and calculates the factorial value of it and prints the Same.

8	Write a Ruby program which counts number of lines in a text files using its regular Expressions facility.
9	Write a Ruby program that uses iterator to find out the length of a string.
10	Write simple Ruby programs that uses arrays in Ruby.
11	Write programs which uses associative arrays concept of Ruby.
12	Write Ruby program which uses Math module to find area of a triangle.
13	Write Ruby program which uses tk module to display a window
14	Define complex class in Ruby and do write methods to carry operations on complex objects.
15	Write a program which illustrates the use of associative arrays in perl.
16	Write perl program takes set names along the command line and prints whether they are regular files or special files
17	Write a perl program to implement UNIX `passed' program
18	An example perl program to connect to a MySQl database table and executing simple commands.
19	Example PHP program for cotactus page.
20	User Authentication: Assume four users user1, user2, user3 and user4 having the passwords pwd1, pwd2, pwd3 and pwd4 respectively. Write a PHP for doing the following. 1. Create a Cookie and add these four user id's and passwords to this Cookie. 2. Read the user id and passwords entered in the Login form (week1) and authenticate with the values (user id and passwords) available in the cookies. If he is a valid user (i.e., user-name and password match) you should welcome him by name (user-name) else you should display "You are not an authenticated user ''. Use init-parameters to do this.
21	Example PHP program for registering users of a website and login.
22	Install a database (Mysql or Oracle). Create a table which should contain at least the following fields: name, password, email-id, phone number (these should hold the data from the registration form). Write a PHP program to connect to that database and extract data from the tables and display them. Experiment with various SQL queries. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page (week2).
23	Write a PHP which does the following job: Insert the details of the 3 or 4 users who register with the web site (week9) by using registration form. Authenticate the user when he submits the login form using the user name and password from the database (similar to week8 instead of cookies).
24	Create tables in the database which contain the details of items (books in our case like Book name , Price, Quantity, Amount) of each category. Modify your catalogue page (week 2)in such a way that you should connect to the database and extract data from the tables and display them in the catalogue page using PHP
25	**HTTP** is a stateless protocol. Session is required to maintain the state. The user may add some items to cart from the catalog page. He can check the cart page

	for the selected items. He may visit the catalogue again and select some more items. Here our interest is the selected items should be added to the old cart rather than a new cart. Multiple users can do the same thing at a time(i.e., from different systems in the LAN using the ip-address instead of local host). This can be achieved through the use of sessions. Every user will have his own session which will be created after his successful login to the website. When the user logs out his session should get invalidated (by using the method session. Invalidate (). Modify your catalogue and cart PHP pages to achieve the above mentioned functionality using sessions.
Text Books:	
1.	Programming the World Wide Web, Robet W Sebesta, 7ed, Pearson.
2.	Web Technologies, Uttam K Roy, Oxford
Reference Books:	
1.	Ruby on Rails Up and Running, Lightning fast Web development, Bruce Tate, Curt Hibbs, Oreilly (2006).
2.	An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, CengageLearning

Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3206	PC	--	--	3	1.5	20	30	3 Hrs.

DATA MINING LAB

(For IT)

Course Objectives:

1	To understand the mathematical basics quickly and covers each and every condition of data mining in order to prepare for real-world problems
2	The various classes of algorithms will be covered to give a foundation to further apply knowledge to dive deeper into the different flavors of algorithms
3	Students should aware of packages and libraries of R and also familiar with functions used in R for visualization
4	To enable students to use R to conduct analytics on large real life datasets
5	To familiarize students with how various statistics like mean median etc

Course Outcomes: At the end of the course, student will be able to

S.No	Outcome	Knowledge Level
1	Extend the functionality of R by using add-on packages	K3
2	Examine data from files and other sources and perform various data manipulation tasks on them	K4
3	Code statistical functions in R	K4
4	Use R Graphics and Tables to visualize results of various statistical operations on data	K3
5	Apply the knowledge of R gained to data Analytics for real life applications	K4

LIST OF EXPERIMENTS

1. Implement all basic Rcommands.
2. Interact data through .csv files (Import from and export to .csvfiles).
3. Get and Clean data using swirl exercises. (Use 'swirl' package, library and install that topic from swirl).
4. Visualize all Statistical measures (Mean, Mode, Median, Range, Inter Quartile Range etc., using Histograms, Boxplots and ScatterPlots).
5. Create a data frame with the followingstructure.

EMP ID	EMP NAME	SALA RY	START DATE
1	Satish	5000	01-11-2013
2	Vani	7500	05-06-2011
3	Ramesh	10000	21-09-1999
4	Praveen	9500	13-09-2005
5	Pallavi	4500	23-10-2000

   a) Extract two column names using columnname.
   b) Extract the first two rows and then allcolumns.
   c) Extract 3^{rd} and 5^{th} row with 2^{nd} and 4^{th} column.
6. Write R Program using 'apply' group of functions to create and apply normalization function on each of the numeric variables/columns of iris dataset to transform theminto
   i) 0 to 1 range with min-maxnormalization.

ii) a value around 0 with z-scorenormalization.

7. Create a data frame with 10 observations and 3 variables and add new rows and columns to it using ‘rbind’ and ‘cbind’function.
8. Write R program to implement linear and multiple regression on ‘mtcars’ dataset to estimate the value of ‘mpg’ variable, with best R^2 and plot the original values in ‘green’ and predicted values in‘red’.
9. Implement k-means clustering using R.
10. Implement k-medoids clustering using R.
11. implement density based clustering on irisdataset.
12. implement decision trees using ‘readingSkills’dataset.
13. Implement decision trees using ‘iris’ dataset using package party and‘rpart’.
14. Use a Corpus() function to create a data corpus then Build a term Matrix and Reveal word frequencies.

Reference Books

1	R and Data Mining: Examples and Case Studies, 1st ed, Yanchang Zhao, Sprnger,2012.
2	R for Everyone, Advanced Analytics and Graphics, 2nd ed, Jared Lander, Pearson,2018.

e-Resources

1	www.r-tutor.com

Code	Category	L	T	P	C	I.M	E.M	Exam
B19IT3207	PR	--	--	--	1	--	50	3 Hrs.
Industrial Training / Skill Development Programmes / Research Project in Higher Learning Institutes								

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3201	MC	3	--	--	--	--	--	--

EMPLOYABILITY SKILLS II

Common to All Branches

Part-A: Verbal and Soft Skills-II

Course Objectives:

1.	To expose the students to bettering sentence expressions and also forming equivalents.
2.	To instill reading and analyzing techniques for better comprehension of written discourses
3.	To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4.	To inculcate conversational skills, nuances required when interacting in different situations
5.	To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes: The students will be able to

S.No	Outcome	Knowledge Level
1	Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.	K3
2	Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences	K3
3.	Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).	K3
4.	Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.	K3
5.	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.	K3

SYLLABUS

UNIT-I	Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT-II	Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author„s intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.
UNIT-III	Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a‗Statement of purpose Letters of Recommendation, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.
UNIT-IV	Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.
UNIT-V	Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1.	Guide to IELTS, Cambridge University Press.
2.	Barron„s GRE guide
3.	Newspapers like‗The Hindu,‗Times of India,‗Economic Times.
4.	Magazines like Frontline, Outlook and Business India.
5.	News channels NDTV, National News, CNN

Text Books:

1.	Objective English and Verbal Reasoning by R S Agarwal
2.	Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP
3.	Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi.

Reference Books:

1.	Oxford Guide to Effective Writing and Speaking by John Seely.
2.	Collins Cobuild English Grammar by Collins
3.	The Art of Public Speaking by Dale Carnegie
4.	The Leader in You by Dale Carnegie
5.	Emotional Intelligence by Daniel Golman
6.	Stay Hungry Stay Foolish by Rashmi Bansal.

Part-B: Quantitative Aptitude-II	
Course Objectives: The objective of introducing quantitative aptitude –II is:	
1.	To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages
2.	To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3.	To familiarize the students with finding out accurate date and time related problems
4.	To enable the students solve the puzzles using logical thinking.
5.	To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating different types of data interpretation problems.	K3
2.	The students will perform efficaciously on analytical and logical problems using various methods.	K3
3.	Students will find the angle measurements of clock problems with the knowledge of calendars and clock.	K3
4.	The students will skillfully solve the puzzle problems like arrangement of different positions.	K3
5.	The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.	K3

SYLLABUS

UNIT-I	Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.
UNIT-II	Puzzle test, blood Relations, permutations, Combinations and probability, Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.
UNIT-III	Periods, Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.
UNIT-IV	Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles

UNIT-V	Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals- Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal. 3. Puzzles to puzzle you by shakunataladevi
3.	Puzzles to puzzle you by shakunataladevi
4.	More puzzles by shakunataladevi
5.	Puzzles by George summers
Reference Books:	
1.	Barron„s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from ‗IMS, Career Launcher and Time„ institutes for competitive exams
4.	Books for CAT by arunsharma
5.	Elementary and Higher algebra by HS Hall and SR knight.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3204	**MC**	**--**	**0**	**3**	**0**	**--**	**--**	**--**

COMPETITIVE CODING

(Common to CSE & IT)

Course Objectives:

1.	To understand the essentials of competitive coding.
2.	To learn the use of Standard Template Library components.
3.	To learn about selection-based algorithms.
4.	To learn different pattern matching and Graph algorithms.

Course Outcomes:

At the end of the course Students will be able to

1.	Use Mathematical functions to solve coding tasks.	K3
2.	Apply STL functions to solve recursive algorithms.	K3
3.	Solve coding tasks related to selection based problems.	K3
4.	Apply Pattern matching and Graph algorithms to solve various problems.	K3
5.	Use Mathematical functions to solve coding tasks.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Competitive Coding** Introduction to Competitive coding and coding Platforms. Coding solution Vs. Efficient Coding solution. Types of solution approaches. Analyzing problem specific data requirement, Various data representations. Essential Data structures for fast coding. Various Syntactical I/O techniques comparison. Numbers, operations (including exponentiation). Integer properties (positive, negative, even, odd, divisible, prime, etc), Fractions, percentages and ratios. Point, vector, Cartesian coordinates (2D integer grid).
UNIT-II (10 Hrs)	**Essentials to Competitive coding** Asymptotic Notations – (Big-O), Evaluating the runtime complexity – Space Complexity - Towers-of-Brahma – Standard Template Libraries - Square root functions, primality testing and related techniques. Euclidean algorithms. Recursion techniques. Organizing data in O(nlogn). Binary search techniques. Basic Techniques: Dynamic Arrays, Set structures, Map structures, Iterators and ranges, Generating Subsets, Generating permutations, Backtracking techniques, Pruning the search. Bit masking.
UNIT-III (10 Hrs)	**Essential Coding Algorithms** Selection based algorithms: sorting, Coin change problem, Fractional selections, Schedules matching, Activity marking, heap sort, Huffman coding techniques, Minimizing sums, Data compression. Finding method count, Subsequence and related problems, paths in grid. DP with Bit mask.

UNIT-IV (10 Hrs)	**String & Tree Algorithms** Naive string searching, z-algorithm, Manacher's algorithm, Rabin-Karp, KMP Algorithm, Suffix arrays. Tree Traversals, Diameter, All longest paths, Applying search property to tree structures. Fenwick tree and Segment Tree.
UNIT-V (10 Hrs)	**Graph Algorithms** Graph Algorithms: Graph Traversals, Shortest paths: Dijkstra's algorithm ,Bellman-Ford Algorithm ,Floyd Warshall Algorithm - Adjacency List Representation – Euler path, tour , cycle – Eulerian Graph - Johnson‟s Algorithm for All-pairs shortest path – Shortest path in Directed Acyclic Graph. Bridges and articulation points. Strongly connected components in directed graphs. 2-SAT.

Reference Books:	
1.	Data Structures and Algorithm Analysis in C++ – Mark Allen Weiss, Third Edition,Pearson Edu Publishers, 2014.
2.	Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, first edition, Tata Mc Graw Hill Publishers.2008.
3.	Advanced Data Structures – Peter Brass, first edition, Cambridge University Press, 2008.
4.	Introduction to Algorithms by ThomosH.cormen, 3rd Edition, PHI,2009.
5.	https://www.geeksforgeeks.org
6.	https://www.tutorialspoint.com